

DAEN/ISEN 427 Homework 1

Due date: Tue Sep 9, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. Explain the difference between a *confounder* and a *collider* in the context of a DAG. Why is it important not to condition on a collider when trying to estimate a causal effect?

Answer:

- (a) Confounders are common causes of both the exposure (A) and outcome (Y).
 - (b) Colliders are common effects (i.e. they are caused by two variables).
 - (c) Conditioning on confounders helps reduce bias, while conditioning on colliders can introduce bias by opening a backdoor path.
2. You are studying whether a new training program (A) improves job performance (Y). Based on domain knowledge, age and prior experience influence both program participation and job performance. Gender is believed to influence only program participation. Construct a DAG using this information, and identify the minimum set of covariates you would need to adjust for to estimate the causal effect of A on Y .

Answer:

- (a) Consider a directed acyclic graph (DAG) with the following structure:
 - $A \rightarrow Y$ (the causal effect we want to estimate)
 - $\text{Age} \rightarrow A$ and $\text{Age} \rightarrow Y$ (confounders are variables that influence both A and Y)
 - $\text{Experience} \rightarrow A$ and $\text{Experience} \rightarrow Y$
 - $\text{Gender} \rightarrow A$ (gender does not affect Y , so it is not a confounder)
 - (b) the minimum set of covariates to adjust for is: Age and Experience (not Gender, affects only A).
This DAG represents the assumed causal relationships among the variables: both Age and Experience directly affect the outcome Y . And the treatment A has a direct causal effect on the outcome Y .
3. Consider a DAG where a treatment variable A causally affects an outcome Y , and there exists a confounder C that influences both A and Y .
 - (a) Explain why the conditional probability $P(Y | A)$ is not equal to the causal effect $P(Y | \text{do}(A))$.
 - (b) Describe how adjusting for C allows us to estimate the causal effect using observational data.

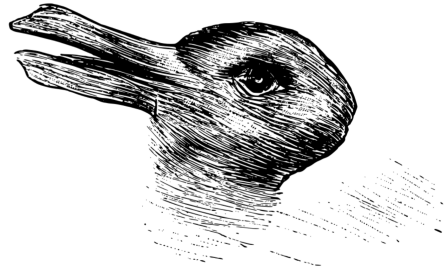
Note: the *do* operator means we are intervening to set A to a value, rather than just observing it.

Answer:

- The expression $P(Y | A)$ includes confounding bias because it captures associations from both causal and non-causal paths (e.g., through C).
- The causal effect $P(Y | \text{do}(A))$ represents the distribution of Y if we were to intervene and set A , breaking the natural causes of A .
Remember, in probability theory $P(Y | A)$ means "the probability of Y given that we observed A ." In causal inference, $P(Y | \text{do}(A))$ means "the probability of Y if we intervene and set A to a specific value." This is a causal statement, not just a statistical one.
- Adjustment for C blocks the backdoor path from A to Y , isolating the causal effect.
- Using the backdoor adjustment formula:

$$P(Y | \text{do}(A)) = \sum_c P(Y | A, C = c) \cdot P(C = c)$$

4. A research article entitled "The Easter bunny in October: Is it disguised as a duck?" explained that "Very little is known about the looks of the Easter bunny on his non-working days." To investigate, the authors showed an "ambiguous drawing of a duck/rabbit...to...265 subjects on Easter Sunday and to 276 different subjects on a Sunday in October of the same year." The authors report: "Whereas on Easter the drawing was significantly more often recognized as a bunny, in October it was considered a bird by most subjects." The drawing shown by the authors in their study was similar to the following figure. Provide a Bayesian perceptual explanation for the authors' results.



Answer: Let us define the perception of the picture by Obs . The two states of the world are $s \in \{\text{"bunny"}, \text{"duck"}\}$. It is reasonable to assume that for a randomly chosen individual, the likelihoods are approximately equal:

$$P(Obs | s = \text{"bunny"}) \approx P(Obs | s = \text{"duck"})$$

However, the prior probability of $s = \text{"bunny"}$ will tend to be higher on a date close to Easter holiday, i.e. $P(s = \text{"bunny"}) > P(s = \text{"duck"})$.

As a result, the posterior

$$\frac{P(s = \text{"bunny"} | Obs)}{P(s = \text{"duck"} | Obs)} = \frac{P(Obs | s = \text{"bunny"})P(s = \text{"bunny"})}{P(Obs | s = \text{"duck"})P(s = \text{"duck"})} \approx \frac{P(s = \text{"bunny"})}{P(s = \text{"duck"})} > 1$$

is larger on a date close to Easter holiday.

5. Suppose that 5% of men and 0.25% of women are color-blind. A person is chosen at random and that person is color-blind. What is the probability that the person is male? (Assume males and females to be in equal numbers.)

Answer: Denote color blind by CB , male by M and female by F . Then

$$\begin{aligned} P(M | CB) &= \frac{P(CB | M)P(M)}{P(CB | M)P(M) + P(CB | F)P(F)} \\ &= \frac{0.05 \times 0.5}{0.05 \times 0.5 + 0.0025 \times 0.5} = 0.9524. \end{aligned}$$

6. Suppose the state of the world s can take the values in $\{s_1, s_2, \dots, s_n\}$. Let x denote an observation and assume $P(x | s = s_i) = P(x | s = s_j)$ $i \neq j$, i.e. the likelihood of observation x is the same for all states of the world. Prove using Bayes' rule the posterior distribution is identical to the prior distribution.

Answer: The posterior distribution satisfies:

$$\begin{aligned} P(s = s_i | x) &= \frac{P(x | s = s_i)P(s = s_i)}{\sum_{j=1}^n P(x | s = s_j)P(s = s_j)}, \quad \text{since } P(x | s = s_j) = P(x | s = s_i) = c \quad \text{then} \\ &= \frac{c P(s = s_i)}{c \sum_{j=1}^n P(s = s_j)}, \quad \text{since } \sum_{j=1}^n P(s = s_j) = 1 \quad \text{then} \\ &= P(s = s_i) \end{aligned}$$